American Board of Toxicology Sample Examination

Key for the following questions:
Choose the single most appropriate answer

(1) Which of the following causes functional alterations in monkeys and man that closely resemble the signs and symptoms of Parkinsonism?

A) lead
B) mercury
C) manganese
D) thallium
E) tellurium

(2) A 16-month-old child ingested an unknown quantity of pills which had been prescribed for his mother. Several hours later he was brought to the emergency room because he could not be aroused. On arrival the child was semicomatose and had labored respiration. The pulse was 160, respiration 30, temperature 38.6° C and blood pressure was 80/50. Gavage produced bright red blood and there was blood in the stool. From these symptoms it is likely that the prescription was

A) headache pills
B) birth control pills
C) iron pills
D) diet pills
E) sleeping pills

(3) The clinical presentation associated with the ingestion of kerosene is usually

A) depression of the central nervous system
B) damage of the liver
C) hydrocarbon pneumonitis
D) ventricular arrhythmia
E) renal damage

(4) Which of the following is the major neurotoxic metabolite of n-hexane?

A) 2-butanone (methyl ethyl ketone)
B) 2,5-hexanedione
C) pentane
D) 2,3 hexanedione
E) none of the above

(5) The most commonly used species in screening for delayed neurotoxic effects of organophosphates is

A) dog
B) chicken
C) cat
D) mouse
E) pig

(6) The pesticide DDT exerts its toxic effects on the nervous system primarily by which of the following mechanisms?

A) blocking acetylcholinesterase at cholinergic synapses
B) inhibiting the release of GABA from nerve endings
C) blocking the efflux of calcium from nerve axons
D) blocking postsynaptic acetylcholine receptors
E) delaying the inactivation of sodium channels

(7) All but one of the following atmospheric pollutants is matched with its probable environmental effect. Which agent is matched to an unlikely environmental consequence?

A) chlorofluorocarbons - stratospheric ozone depletion
B) carbon monoxide - spruce forest die-off
C) nitrogen oxides - photochemical smog
D) carbon dioxide - global warming
E) sulfur dioxide - acid precipitation

(8) Chlorine as a gas causes

A) simple asphyxiation
B) methemoglobinemia
C) carboxyhemoglobinemia
D) cholestasis
E) pulmonary edema

(9) A glass bottle containing one curie of (^{3}H) cholesterol is left on the table in the center of a laboratory of 1000 square feet in which several people work. What would most likely be the effect on the workers after two weeks?

A) no effect
B) moderate depression of the lymphocyte count with increased incidence of infections
C) moderate gastrointestinal upset by the end of the two-week period
D) death of at least half of the workers due to acute radiation sickness
E) an increased incidence of malignant tumors after a latent period of about 10 years

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(20) The mechanism by which a sperm contains both members of a chromosomal pair is
A) nondisjunction
B) clastogenesis
C) mitosis
D) mutagenesis
E) apoptosis

(21) Which of the following would be appropriate to use as a positive control for S9 metabolic activation in *a Salmonella typhimurium* (Ames) assay?
A) sodium azide
B) mitomycin C
C) nitrogen mustard
D) 2-aminoanthracene
E) all of the above

(22) The increase in teratogenicity of salicylate when coadministered with sodium benzoate in the rat is directly related to
A) increased fetal blood volume
B) decreased metabolism of salicylate
C) decreased plasma protein binding of salicylate
D) increased maternal blood volume
E) increased excretion of water-soluble metabolites

(23) A majority of congenital malformations in humans is due to
A) drugs
B) emotional stress
C) environmental pollutants
D) infectious agents
E) unknown causes

(24) Many inflammatory mediators in the lung are released from cells participating in the inflammatory process. Which of the following mediators is NOT released by the macrophage/ monocyte?
A) complement factors
B) interleukin-1
C) toxic oxygen species
D) histamine
E) tumor necrosis factor (TNF)

(25) Which of the following diseases of the lung is NOT properly paired with its cause?
A) farmer's lung caused by spores of thermophilic actinomyces
B) bagassosis caused by dust from dried sugar cane
C) byssinosis caused by cotton flax or hemp dust
D) asthma-like symptoms caused by toluene diisocyanate
E) cheese washer's lung caused by exposure to toxin eaten by cattle and concentrated in their milk

(26) Thiamine is useful in the treatment of peripheral neuropathies induced by which of the following?
A) arsenic and ethanol
B) ethanol and carbon dioxide
C) carbon dioxide and carbon disulfide
D) ethanol and carbon disulfide
E) all of the above

(27) Ginger jake contaminated ginger extract was caused by
A) deltamethrin
B) methanol
C) triorthocresyl phosphate (TOCP)
D) triethyltin
E) acrylamide

(28) The most reliable animal model, to date, for the study of immunologic reactions of the delayed hypersensitivity type in contact photosensitivity dermatitis is the
A) rat
B) rabbit
C) guinea pig
D) gerbil
E) monkey

(29) Evidence for the allergic nature of photosensitization was obtained by
A) irradiation of active compounds before application to test animals
B) passive transfer of contact photoallergy from actively sensitized to non-sensitized guinea pigs
C) irradiation of test animals before application of active compounds
D) isolation of the photoantigen
E) correspondence of inducing wavelengths within absorption spectra of active compounds

Key for the following questions:
Choose the single most appropriate answer

(39) Damage to proximal convoluted tubules in the nephrons following exposure to inorganic mercury or uranium might be expected to be reflected in which of the following?

A) aminoaciduria
B) elevated aspartate aminotransferase (AST)
C) decreased glomerular filtration rate (GFR)
D) proteinuria
E) elevated blood glucose

(40) In order to properly select three dose levels for a subchronic toxicity study, which of the following factors is least important to know prior to the study

A) potential of the chemical to bioaccumulate
B) shape of the dose-response curve from acute studies
C) target organs
D) linearity of pharmacokinetics with dose

(41) The biotransformation of relatively inert chemicals to highly reactive metabolites (metabolic activation) is an important component of chemical-induced toxicities. It is true of this process that

A) phase I reactions (oxidations, reductions) generate reactive but protective metabolites
B) phase II reactions (conjugations) generate reactive toxic metabolites
C) a relatively inert chemical is usually harmless but once biotransformation occurs the end result is enhanced toxicity
D) all of the above
E) none of the above

(42) The initial oxidant species formed when an electron is transfered to molecular oxygen is

A) singlet oxygen
B) superoxide anion
C) hydroxyl radical
D) hydrogen peroxide
E) hydronium radical

(43) If the rate of excretion of a chemical is 0.3 Gm per hour, which of the following is true?

A) its biological half-life is 2.31 hours
B) it follows first order kinetics
C) it follows zero order kinetics
D) 30% of the chemical is excreted in an hour
E) none of the above

(44) A toxicant would be expected to accumulate in the body during a period of repeated exposure when the

A) volume of distribution is large
B) half-life is shorter than the exposure interval
C) half-life is longer than the exposure interval
D) toxicant is absorbed into soft tissues
E) rate of absorption is greater than the rate of excretion

(45) Which of the following microsomal inducers increase(s) bile flow and excretion?

A) benzo(a)pyrene
B) 3-methylcholanthrene
C) phenobarbital
D) all of the above
E) none of the above

(46) An example of the ability of genetic factors to influence the susceptibility to toxic agents is

A) genetic variation of glucuronyl transferase levels in cats
B) differences in the ability of dogs to acetylate arylamines
C) the sulfonamides' displacement of bilirubin from its plasma-binding sites in premature infants
D) hemolytic effects of primaquine in individuals with glucose-6-phosphate dehydrogenase deficiency
E) nephrotoxic effects of chloroform in mice

(47) Which of the following reactions is NOT inhibited by piperonyl butoxide?

A) aminopyrine N-demethylation
B) parathion desulfuration
C) aflatoxin B1 epoxidation
D) naphthalene hydroxylation
E) malathion hydrolysis

Key for the following questions:
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(57) Which of the following air pollutants is not classified as a pulmonary irritant?
A) nitrogen dioxide
B) carbon monoxide
C) sulfur dioxide
D) formaldehyde

(58) Which of the following has NOT been implicated in causing metal fume fever?
A) aluminum
B) beryllium
C) copper
D) iron
E) tin

(59) All of the following statements concerning the biologic assessment of exposure are true EXCEPT
A) biologic monitoring of exposure includes an estimation of the internal dose
B) monitoring the airborne concentration of an industrial pollutant provides a reliable estimate of the amount absorbed by a worker
C) the concentration of polychlorinated biphenyls in blood is proportional to the amount accumulated in fatty tissue
D) biologic monitoring of exposure is of practical value only when relationships between external exposure, internal dose and adverse effects are known

(60) A wide spectrum of lung changes varying from a clinically insignificant focal pigmentation to crippling fibrosis is most likely to be encountered among which of the following?
A) distillery workers
B) coal miners
C) farmers
D) histologists
E) city traffic directors

(61) The use of British Anti-Lewisite (BAL) to treat poisoning by arsenic is an example of
A) functional antagonism
B) receptor antagonism
C) chemical antagonism
D) dispositional antagonism

(62) Renal papillary necrosis (RPN) has been associated with exposure to
A) mixtures of aspirin and phenacetin
B) aspirin
C) acetaminophen
D) nsaids (non-steroidal antiinflammatory drugs)
E) all of the above

(63) Opisthotonus, which may occur following a toxic dose of strychnine, refers to
A) paralysis of involuntary muscles
B) labored breathing
C) back arched in hyperextension with only the head and feet touching the ground
D) muscles of the face and neck relaxed
E) all of the above

Key for the following questions:
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

(64) Heavy metal antagonists
1) react with heavy metals to form tightly bound complexes
2) react with heavy metal to form chelates
3) act by preventing the binding of toxic metals to body ligands
4) depend solely upon ring size of the resulting complex for chelate stability

(65) BAL (British anti-lewisite) is used in the treatment of
1) cadmium toxicity
2) selenium toxicity
3) tellurium toxicity
4) arsenic toxicity

(66) Benzene metabolism
1) does not lead to opening of the aromatic ring
2) results only in urinary phenol and catechol excretion
3) is affected by urinary pH
4) involves cytochrome P450, epoxide hydratase and glutathione S-epoxide transferase

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(76) Which of the following groups of drugs is/are correctly paired with the predominant toxic side effect?

1) beta adrenergic blocking agents (e.g., propranolol) - hypotension and bradycardia
2) classical antihistamines (e.g., diphenhydramine) - sedation
3) aminoglycoside antibiotics (e.g., gentamicin) - ototoxicity and renal toxicity
4) nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) - gastrointestinal irritation

All E

(77) Cytological examinations are useful in the detection of the following types of mutations:

1) breaks
2) base-pair substitution
3) deletions
4) frameshift

Aberrations

(78) Test(s) to detect chromosomal mutations

1) Ames *salmonella* tests
2) dominant lethal test in mammals
3) specific locus test in mice
4) *in vivo* cytogenetic tests

(79) Which of the following are true of the dominant lethal assay?

1) both the male and female are exposed to the test chemical prior to mating
2) exposure of the female to the test chemical continues throughout gestation
3) dominant lethals are calculated as the ratio of dead-born fetuses to total fetuses
4) test designed to detect mutagenic effects in the spermatogenic process that are lethal to the embryo or fetus

mates only

(80) Which of the following is/are true integrative processes of the hypothalamo-pituitary-gonadal axis?

1) FSH and LH are glycoproteins synthesized and released from a subpopulation of the basophilic gondadotropic cells of the pituitary gland
2) luteinizing hormone-releasing hormone (LH-RH) acts on gonadotropic cells, thereby stimulating the release of FSH and LH
3) LH-RH and follicle-stimulating hormone-releasing hormone (FSH-RH) appear to be the same substance
4) gonadotropins are not important in the female

A

(81) It is currently claimed that primary lung cancer is

1) caused by heavy cigarette smoking over a period of years
2) decreasing in incidence in both males and females
3) related etiologically to air pollution
4) highly predictable on the basis of genetics

B

(82) The theoretical time for a test compound gas to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the t99. In addition to the equation constant what measurements are needed to calculate the t99?

1) chamber volume
2) chamber target concentration of the test compound
3) chamber airflow rate
4) molecular weight of the test compound

B

(83) Peripheral axonopathies may be caused as a result of overexposure to which of the following agents?

1) acrylamide - gels
2) taxol - microtubules
3) ethyl alcohol -
4) 3,3'-iminodipropionitrile - IDPN - (TOCP) waltzing syndrome

E

1,2,3 WRT 2
1,2,3

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(93) In nitrite poisoning
1) hemoglobin is oxidized to methemoglobin
2) the blood has a brown color
3) methemoglobin is ineffective as an oxygen carrier
4) methylene blue is an effective antidote because it is a reducing agent which rapidly converts methemoglobin to hemoglobin

(94) Increase the concentration of carbon monoxide in the blood by more than one percent
1) methylene chloride
2) smoking
3) hemolysis
4) automobile exhaust

(95) Which of the following statements concerning the kidney is/are true?
1) phenobarbital and salicylates will be cleared more rapidly from the plasma after IV administration of sodium bicarbonate
2) amphetamines will be cleared more rapidly from the plasma after IV administration of ammonium chloride
3) the mechanism for secretion of organic compounds is localized in the proximal tubule
4) because of high blood flow to the medulla, it is very sensitive to toxic insult

(96) The effect of toxic chemicals on renal function can be assessed by measurement of
1) inulin clearance
2) blood urea nitrogen
3) urine protein concentration
4) plasma creatinine

(97) Toxicity due to synergism between two drugs can most likely occur if the drugs
1) have similar pharmacologic activity
2) are toxic at high dose levels
3) have the same target organs
4) interact to mutually inhibit absorption across membranes

(98) Which of the following statements are characteristic for both resistance and tolerance?
1) effects are readily reversible
2) result from natural selection processes in the group
3) are due to genetic factors (inherited)
4) always result in a decrease in the toxic response

(99) Knowing that a drug is a base, pKa=10, and that its nonionized form has a low lipid solubility could enable one to predict that the drug would
1) not readily cross the placental barrier
2) have a small volume of distribution
3) penetrate the CNS poorly
4) have a long biological half-life ($t_{1/2}$)

(100) Which of the following agents have been useful in studying mechanisms associated with oxygen radical induced injury either *in vivo* or *in vitro*?
1) superoxide dismutase
2) hydroxyl radical scavengers
3) metal chelators
4) antioxidants

(101) The Monte Carlo procedure
1) is a statistical method used to distribute animals among dose groups and eliminate bias
2) results in a description of the probability distribution of risk
3) is most appropriate when group sizes are large, such as those found in carcinogenicity studies
4) is a statistical method which randomly selects values for each of many variables used as input for risk calculations

(102) Quantitaive cancer risk estimation, based on results of animal toxicity tests
1) is based on the results obtained at the maximum tolerated dose
2) is extrapolated only using linear dose-response relationships
3) is based on application of multiple safety factors
4) is based on both statistical procedures and mechanistic models of chemical carcinogenesis